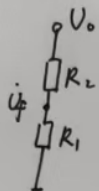
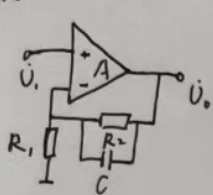


Homework 9.

1. (1) 既有直流也有交流反馈. 电压串联负反馈.

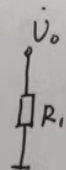
直流:



$$F = \frac{U_f}{U_o} = \frac{I_o \cdot R_1}{I_o \cdot (R_1 + R_2)} = \frac{R_1}{R_1 + R_2}$$

$$A_{uf} = \frac{1}{F} = 1 + \frac{R_2}{R_1}$$

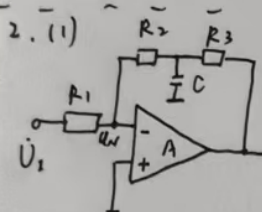
交流:



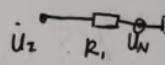
$$F = \frac{U_f}{U_o} = 1$$

$$A_{uf} = \frac{1}{F} = 1$$

(2) 同(1) 交流直流.

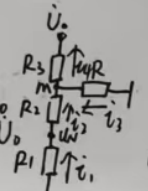
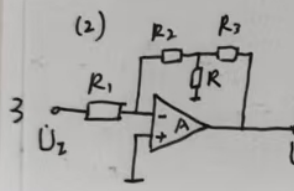


只有直流反馈: 电压并联负反馈.



$$F = \frac{U_f}{U_o} = \frac{U_N - U_o}{R_1} \quad \text{令 } U_N = 0 \quad \text{则 } F_{in} = \frac{i_{R1}}{U_o} = -\frac{1}{R_1} = -\frac{1}{R_2 + R_3}$$

$$A_{uf} = \frac{U_o}{U_i} \approx \frac{U_o}{i_{R1} R_1} \approx \frac{U_o}{i_{R1} R_1} = \frac{1}{F} \cdot \frac{1}{R_1} = -\frac{R_2 + R_3}{R_1}$$



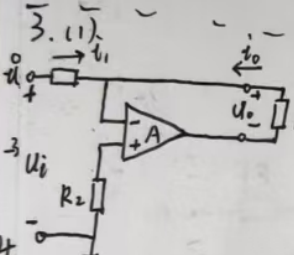
$$i_2 = i_1 = \frac{U_i}{R_1}$$

$$U_N = -\frac{R_2}{R_1} U_i$$

$$U_o = U_N - (i_2 + i_3) \cdot R_4$$

$$\text{则 } A_{uf} = \frac{U_o}{U_i} = -\frac{R_2 + R_4}{R_1} \cdot \frac{R_4 + R_3}{R_3} \cdot \frac{1}{U_i}$$

$$F = \frac{U_f}{U_o} = \frac{1}{U_o} \cdot \frac{0 - U_o}{(R_2 + R_4) + R_3} = -\frac{1}{(R_2 + R_4) + R_3}$$



电流并联负反馈.

$$U_p = U_N = 0 \quad \text{则 } i_1 + i_2 = 0$$

$$\frac{U_i}{R_1} + \frac{U_o}{R_L} = 0$$

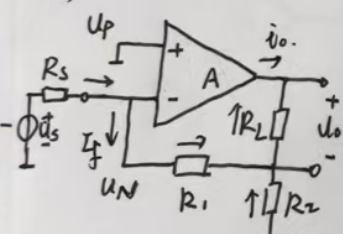
$$F_{in} = \frac{i_{R1}}{U_i} = -\frac{1}{R_1}$$

$$i_1 = \frac{U_i - U_N}{R_1} = \frac{U_i}{R_1}$$

$$\text{则 } A_{uf} = \frac{U_o}{U_i} = -\frac{R_L}{R_1}$$

$$i_2 = \frac{U_o - U_N}{R_L} = \frac{U_o}{R_L}$$

(2)

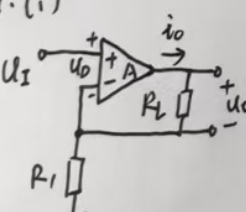


$$F_{ii} = \frac{I_f}{I_o} = -\frac{R_2}{R_1 + R_2}$$

$$U_s \approx I_f R_s \quad U_o = I_o R_L$$

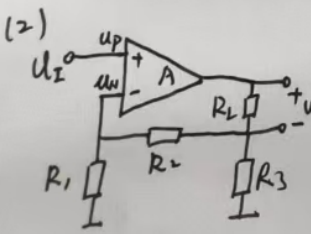
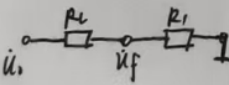
$$A_{uf} = \frac{U_o}{U_s} \approx \frac{1}{F_{ii}} \cdot \frac{R_L}{R_s} = -(1 + \frac{R_1}{R_2}) \frac{R_L}{R_s}$$

4. (1) 电流串联负反馈.



$$F = \frac{U_f}{U_i} = R_1$$

$$A_{uf} = \frac{U_o}{U_i} \approx \frac{i_o \cdot R_L}{U_i} = \frac{R_L}{R_1}$$



$$F = \frac{U_f}{U_o} = \frac{R_1 R_3}{R_1 + R_2 + R_3} \quad U_N = U_p = U_i, \quad i_{R2} = i_{R1} = \frac{U_i}{R_1}$$

$$A_{uf} = \frac{i_o R_L}{U_i} = \frac{R_1 + R_2 + R_3}{R_1 R_3} \cdot R_L$$

$$i_{R3} = \frac{U_{R1} + U_{R2}}{R_3} = (1 + \frac{R_2}{R_1}) U_i / R_3$$

$$i_o = i_{R2} + i_{R3} = \frac{R_1 + R_2 + R_3}{R_1 R_3} \cdot U_i$$